Benedict Thomas R

Mechanical Engineering Student | Product Design & Fabrication
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Portfolio
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SUMMARY

I'm a third-year Mechanical Engineering student with hands-on experience in SolidWorks CAD, MIG welding, and fabrication through active involvement in an SAE India EV BAJA team. Comfortable working across the design-to-build cycle — from structural modelling and engineering drawings to shop-floor assembly. Supplementary exposure to embedded systems and electronics through independent hardware projects.

EDUCATION

Loyola-ICAM College of Engineering and Technology *2024-28*

B.E. Mechanical Engineering | GPA: 8.00 / 10.0

EXPERIENCE

Design & Manufacturing Team Member · EV BAJA — SAE India *August 2024-Present*

- Designed secondary structural members in SolidWorks, maintaining load-path integrity.
- Interpreted engineering drawings on the shop floor to guide fabrication through tube cutting, jig-based alignment, and MIG welding.
- Contributed to full vehicle assembly, ensuring integration of manufactured components with drivetrain and suspension systems.

PROJECTS

Compact Reaction Wheel for Drone Attitude Control — *TN IMPACT Hackathon 2026* *2026*

- Designed a 3-spoke aluminium reaction wheel in SolidWorks with full engineering drawing package for a compact drone stabilisation system.
- Integrated wheel with BLDC motor, MPU-6050 IMU, L298N motor driver, and ESP32 microcontroller into a functional hardware demonstrator.
- Validated attitude-control response through bench testing, demonstrating stable closed-loop feedback from IMU sensor data.

Trash Can Compactor Redesign — *Project-Driven Learning* *2026*

- Identified core UX failure in the Joseph Joseph Titan compactor — upward bag removal requires lifting 10+ kg overhead, causing musculoskeletal strain.
- Redesigned with a front cabinet door using a hole-pin hinge and push-to-open latch, enabling horizontal single-hand bag removal.
- Reinforced door with internal ribs sized to distribute compaction loads and maintain latch alignment across repeated cycling.

SKILLS

CAD / Software: SolidWorks (part & assembly modelling, engineering drawings)
Fabrication: MIG welding, jig-based alignment, tube cutting, manual assembly, sheet metal
Design: Conceptual sketching, design-for-manufacturing, engineering drawing interpretation
Languages: English (fluent), Tamil (native), Deutsch (A1)

CERTIFICATIONS & VIRTUAL EXPERIENCE

Siemens Operations Industrial Engineer Virtual Experience (Forage, 2025) — time studies, factory layout proposals
GE Aerospace Explore Engineering Virtual Experience (Forage, 2025) — energy source design, compression/bypass ratio analysis